



pAllette

Your Creativity, AI Support
Option A
Final Checkpoint

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Problem & Motivation

Digital painting tools give users full control, but they often provide limited support when users know the mood or concept they want without knowing the exact color or next creative step.

This creates friction, because users must interrupt their flow to search, test, and adjust instead of staying focused on drawing.

Our goal was to design an AI-enabled system that supports creativity without taking control away from the user, in line with the course focus on human-AI interaction rather than algorithmic complexity.

- Users think in concepts, moods, and intentions.
- Creative flow breaks when support is external to the tool.
- AI should assist the process, not replace the artist.

What pAllette is Now

pAllette is now an AI-augmented drawing environment rather than only a color suggestion concept.

The current prototype combines a digital painting workspace with several AI-assisted interactions inside the same interface.

This allows the system to support the user at different stages of the creative workflow, from choosing colors to generating and continuing sketches.

- Integrated drawing workspace
- Multiple AI-assisted interaction modes
- Human remains in control of the outcome

Implemented AI Models

The final prototype includes several AI-related modules that operate inside the same drawing application. These modules are not equally mature, but together they demonstrate a broader view of human-AI collaboration in creative tools.

We use a mixed-initiative system in which the AI proposes suggestions, and the human evaluates, accepts, edits, or ignores the result, retaining final control.

These are the current features our project offers in this final checkpoint.

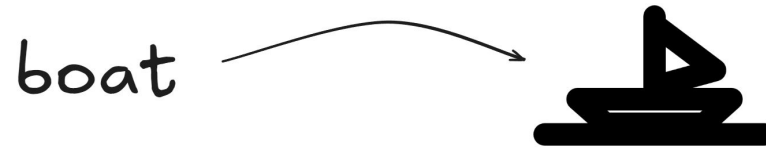
- Palette-aware color and feature suggestion
- Prompt-to-sketch support
- Webcam and canvas exploration by doodle
- SketchRNN-based drawing completion

Main Interaction Patterns

1. Prompt / doodle / webcam → color suggestions



2. Prompt / doodle → sketch support

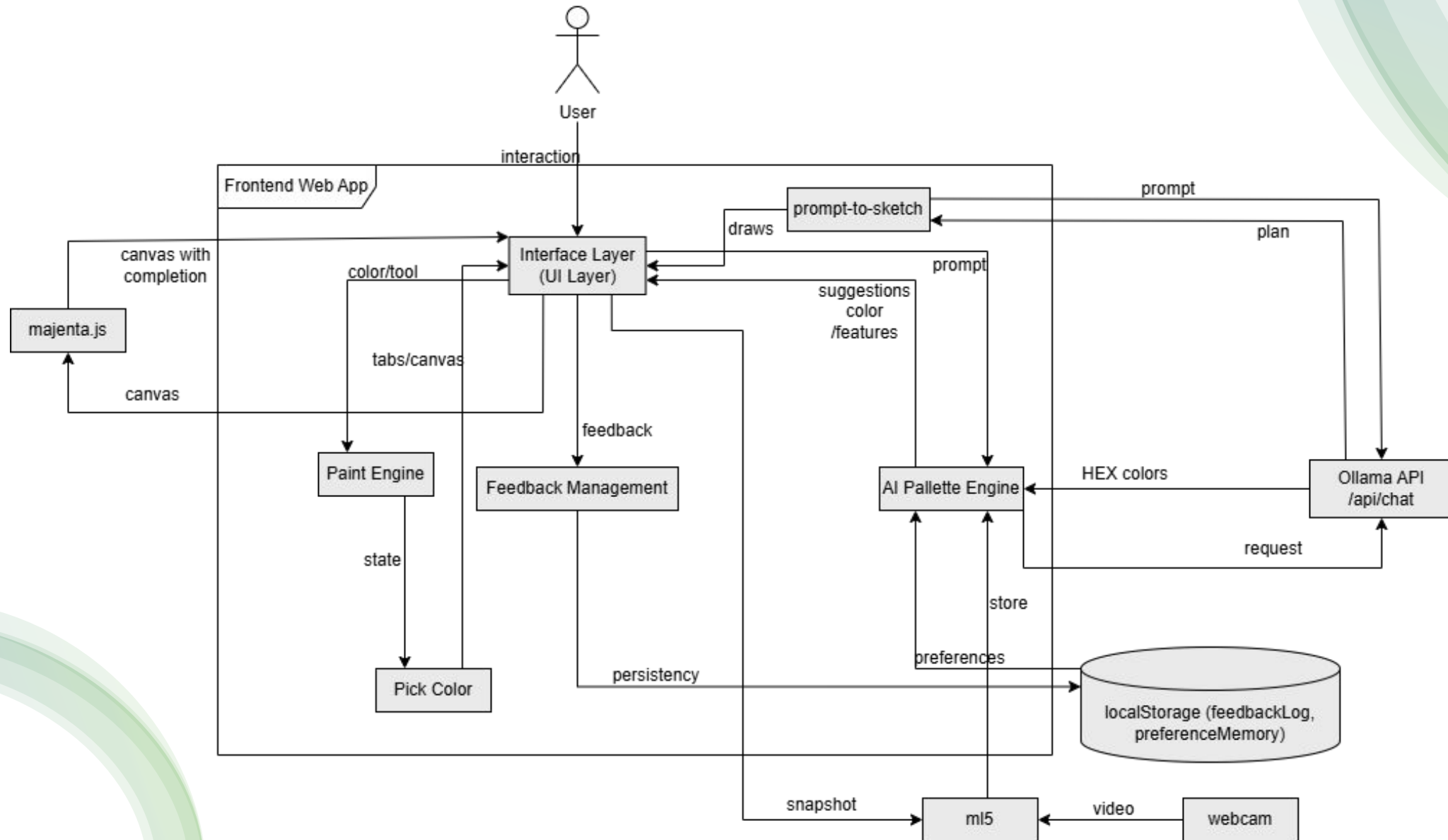


3. Existing drawing → AI completion



System Architecture

System Architecture - pAlIette



The system collects user input through prompts, drawing actions, color choices, and interaction decisions made inside the interface.

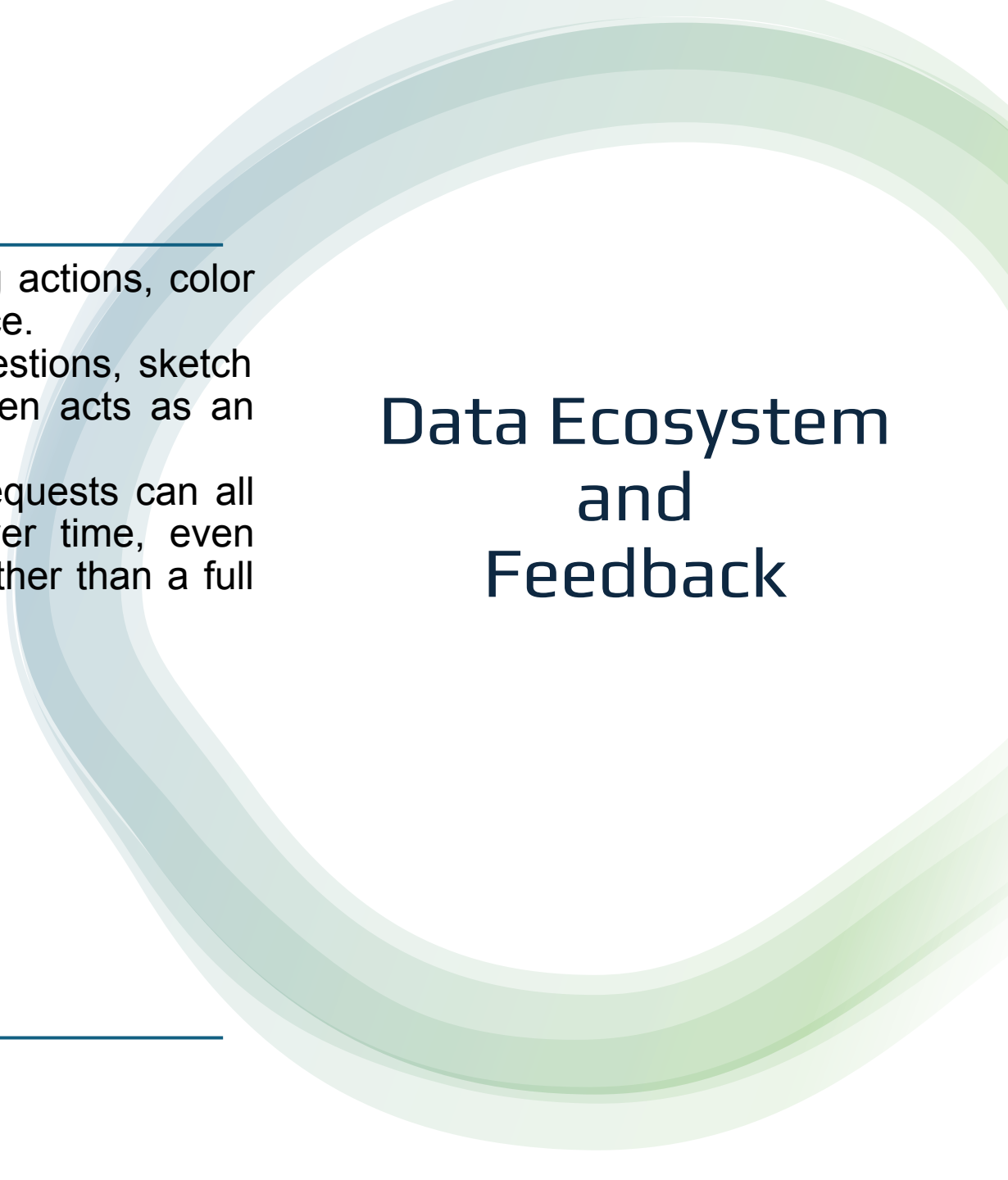
The AI components process this input to generate suggestions, sketch outputs, or drawing continuations, and user behavior then acts as an implicit feedback signal.

Accepted suggestions, rejected outputs, and repeated requests can all help describe how the user influences the system over time, even though the current prototype is still a proof of concept rather than a full learning pipeline.

Input: prompts, drawing state, user actions

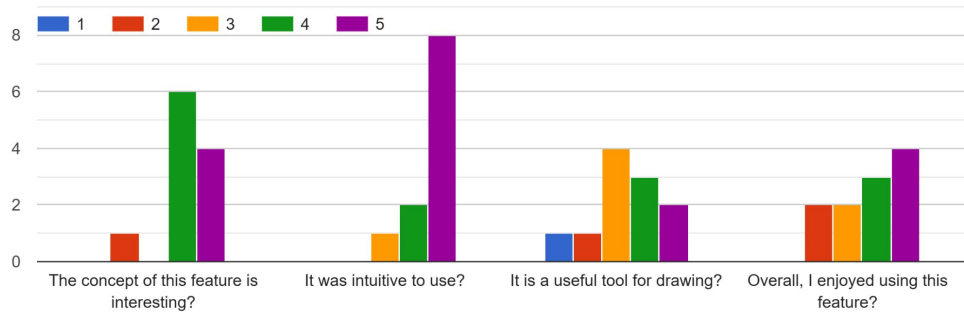
Output: colors, sketches, continuations

Feedback: accept, reject, refine, regenerate

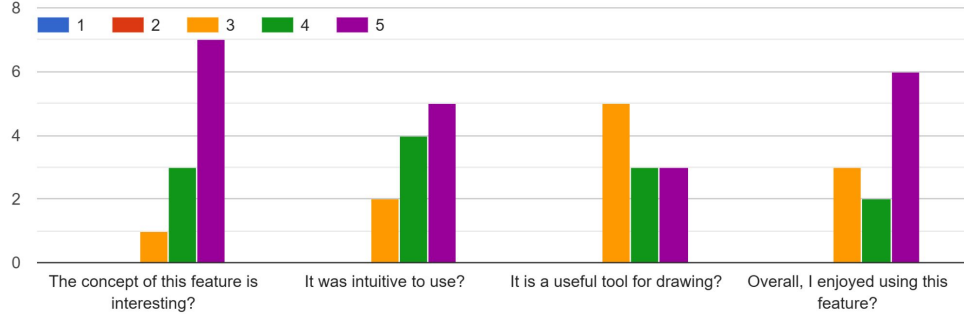


Data Ecosystem and Feedback

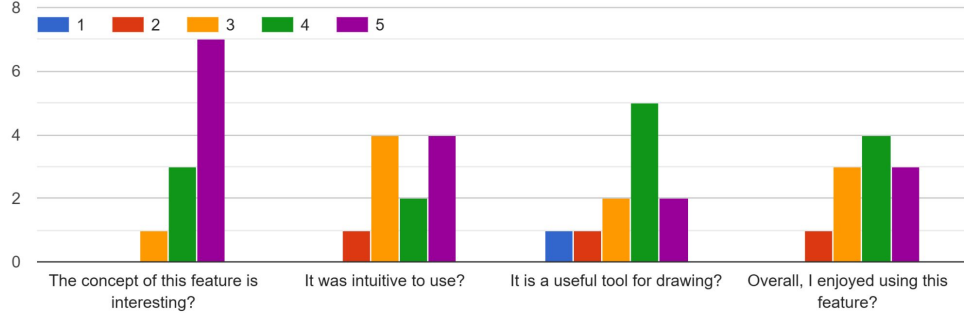
In scale between 1 (least) and 5 (most), what's your opinion about the feature sketch prompt



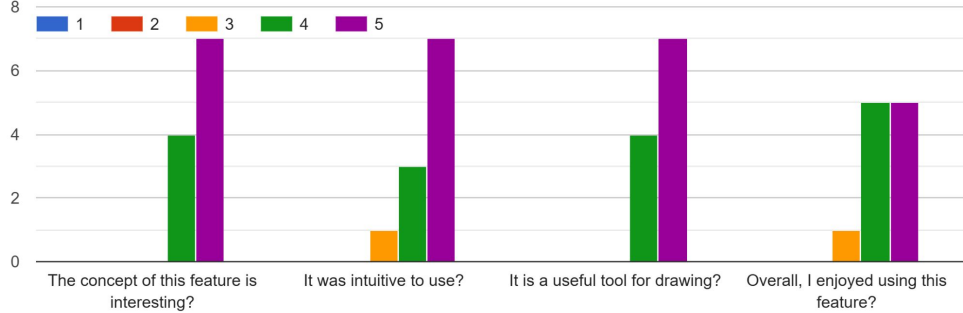
In scale between 1 (least) and 5 (most), what's your opinion about the feature doodle canvas



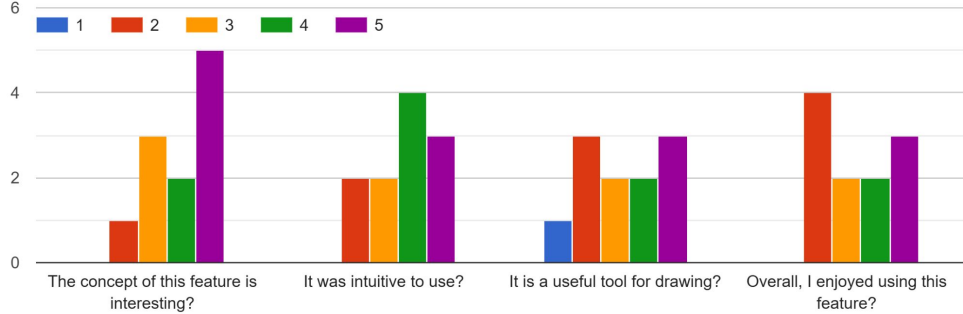
In scale between 1 (least) and 5 (most), what's your opinion about the feature webcam



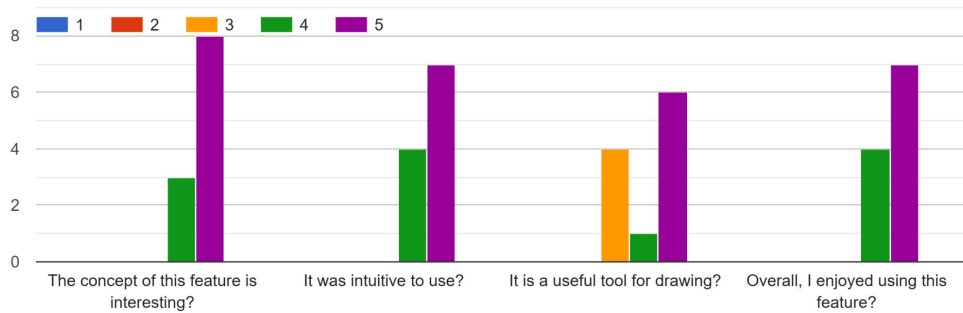
In scale between 1 (least) and 5 (most), what's your opinion about the feature colour by prompt



In scale between 1 (least) and 5 (most), what's your opinion about the feature AI autocomplete



In scale between 1 (least) and 5 (most), what's your opinion about all of the paint system?



Feedback

Limitations & Challenges

- ❖ Ambiguity in prompts and creative intent
- ❖ Hardware and model limitations
- ❖ Uneven maturity across modules

Future Work

- ❖ Personalization from interaction history
- ❖ Implementation of feedback for all human-AI interactions
- ❖ Stronger multimodal understanding of the canvas
- ❖ More user study on creative flow and usability

Conclusion

pAllette explores how AI can support creativity through an integrated drawing environment that combines color assistance, sketch support, and drawing continuation. Rather than automating the creative process, the project focuses on reducing friction, expanding exploration, and keeping the user in control of decisions and outcomes. This makes the project a practical example of human-centered AI design, aligned with the goals of the course and grounded in a working prototype.

- AI support without loss of user agency
- Broader creative assistance in one interface
- Final prototype demonstrates the concept in practice